



Design Challenge Report

Rail-to-Rail Class-AB Folded-Cascode Op-Amp

Analog IC Design | Cadence Virtuoso 65nm

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Technology	TSMC 65 nm CMOS (2.5 V Devices)	Supply Voltage (V_{DD})	2.5 V
Quiescent Output Current (I_Q)	200 μ A	Input Stage	Complementary Rail-to-Rail
Tail/Bias Branch Budgets	$I_{b1,2,5,8} = 10 \mu$ A, $I_{b6,7} = 40 \mu$ A	Output Stage Topology	Monticelli Translinear Class-AB
Nominal Load Network	$R_L = 10 \text{ k}\Omega$ to V_{MID} , $C_L = 20 \text{ pF}$	Small-Signal Targets	$A_{OL} > 60 \text{ dB}$, $PM > 60^\circ$, $f_u > 10 \text{ MHz}$

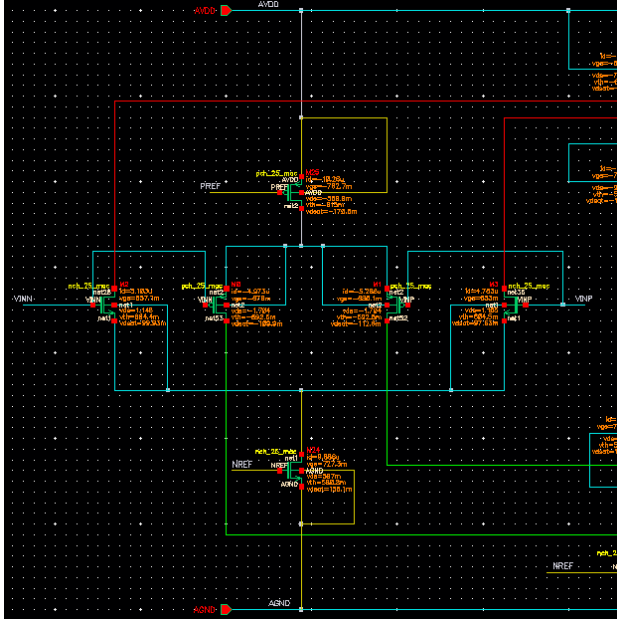
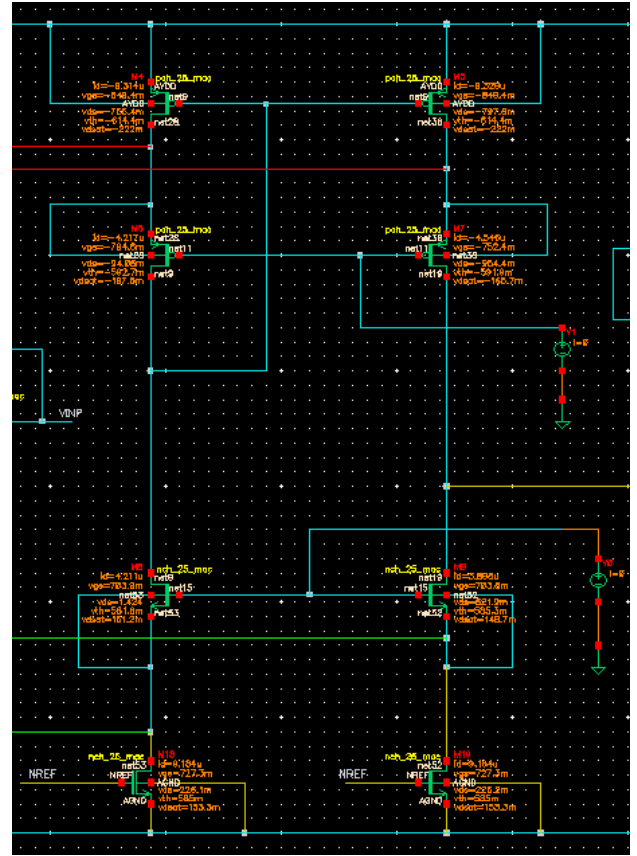
Table 1. Target design parameters and performance constraints.

Deliverable 1: Closed-Loop DC Operating Point Analysis

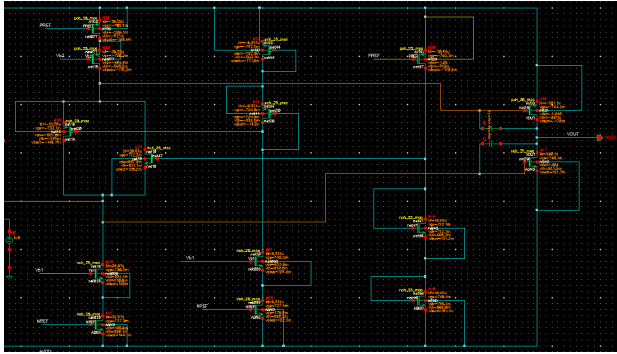
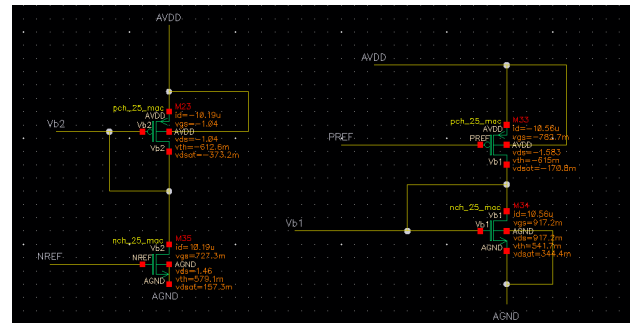
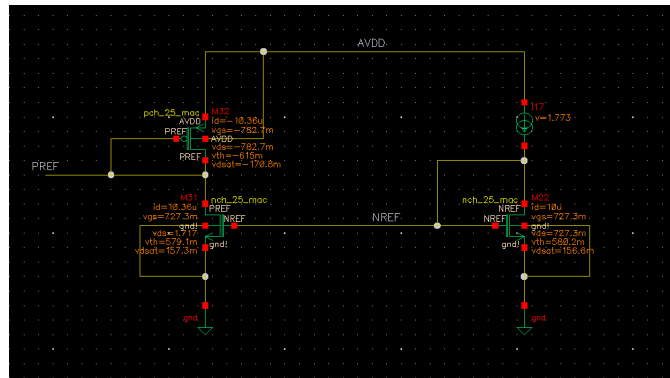
Design Challenge Objective

Evaluate the closed-loop unity-gain buffer configuration (TB1) under nominal input common-mode excitation ($V_{ICM} = V_{MID} = 1.25 \text{ V}$). Display annotated schematic captures confirming device operating points ($I_D, V_{GS}, V_{DS}, V_{dsat}$) and verify that every transistor in the core resides strictly within the saturation region (region = 2).

1. Annotated Subcircuit Operating Point Schematics

(a) Rail-to-rail input stage and tail sources (I_{b1} , I_{b2}).

(b) Folded-cascode stage and active current-mirror loads.

(c) Class-AB Monticelli loop and 200 μ A output stage.(d) Self-tracking cascode bias generators (V_{b1} , V_{b2}).

(e) Master reference seed and current mirror distribution.

Figure 1. Deliverable 1: Virtuoso DC operating point captures across all functional blocks in TB1 ($V_{DD} = 2.5$ V, $V_{ICM} = 1.25$ V, $R_L = 10$ k Ω to V_{MID}).

2. Operating Point Tabulation & Headroom Verification

Table 2 summarizes the operating parameters extracted from the simulation annotations. The output push-pull devices (M16, M17) lock at $I_Q \approx 200 \mu\text{A}$, the cascode branches carry $40 \mu\text{A}$, and the folded cascode operates symmetrically with $\approx 5 \mu\text{A}$ per input/cascode branch. All active devices satisfy $|V_{DS}| > |V_{dsat}|$ (region = 2).

Device / Stage	Type	$I_D [\mu\text{A}]$	$ V_{GS} [\text{mV}]$	$ V_{DS} [\text{mV}]$	$ V_{dsat} [\text{mV}]$	Region
Input Stage & Tail Sources						
M25 (I_{b1} Tail)	PMOS	-10.28	782.7	308.8	170.8	2 (Sat)
M24 (I_{b2} Tail)	NMOS	9.88	727.3	387.0	138.1	2 (Sat)
M0, M1 (PMOS Diff)	PMOS	-4.97 / -5.28	878.0 / 830.1	1704	109.9 / 112.8	2 (Sat)
M2, M3 (NMOS Diff)	NMOS	5.10 / 4.78	837.7 / 833.0	1148	99.3 / 97.6	2 (Sat)
Folded-Cascode Stage						
M4, M5 (PMOS Load)	PMOS	-8.31 / -8.33	848.4	766.4 / 797.9	222.0	2 (Sat)
M6, M7 (PMOS Cascode)	PMOS	-4.22 / -4.35	784.6 / 752.4	94.06 / 904.4	187.6 / 168.7	2 (Sat)
M8, M9 (NMOS Cascode)	NMOS	4.21 / 3.89	703.8	1424 / 621.9	161.2 / 148.7	2 (Sat)
M18, M19 ($I_{b3,b4}$)	NMOS	9.18	727.3	226.1 / 226.2	153.3	2 (Sat)
Monticelli Floating Bias & Output Stage						
M29, M30 (I_{b7})	PMOS	-39.22	782.7 / 770.3	269.3 / 494.9	169.4 / 175.2	2 (Sat)
M27, M28 (I_{b6})	NMOS	39.87	727.3 / 750.3	166.9 / 581.1	144.1 / 155.0	2 (Sat)
M10, M11 (Translinear)	PMOS/NMOS	-18.59 / 20.83	733.1 / 712.2	827.8 / 807.0	148.7 / 128.2	2 (Sat)
M16 (Output PMOS)	PMOS	-201.1	754.2	1249	170.0	2 (Sat)
M17 (Output NMOS)	NMOS	200.7	748.1	1254	151.7	2 (Sat)
Master References & Bias Generators						
M22, M32 (NREF/PREF)	NMOS/PMOS	10.00 / -10.36	727.3 / 782.7	727.3 / 782.7	156.6 / 170.8	2 (Sat)
M23, M35 (V_{b2} Gen)	PMOS/NMOS	-10.19 / 10.19	1040 / 727.3	1040 / 1460	373.2 / 157.3	2 (Sat)
M33, M34 (V_{b1} Gen)	PMOS/NMOS	-10.56 / 10.56	782.7 / 917.2	1583 / 917.2	170.8 / 344.4	2 (Sat)

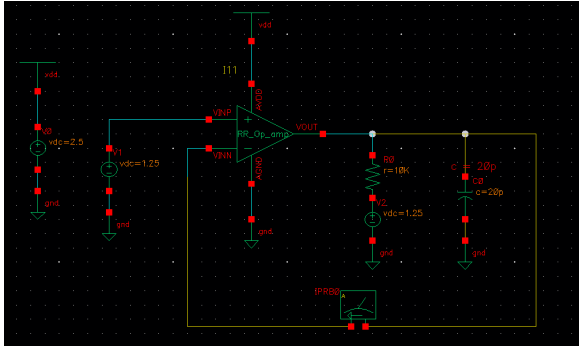
Table 2. Deliverable 1: Consolidated simulated DC operating point parameters demonstrating proper biasing across all operational amplifier branches in closed-loop unity gain ($V_{ICM} = 1.25 \text{ V}$).

Deliverable 2: Small-Signal Stability Analysis (stb)

Design Challenge Objective

Evaluate the small-signal open-loop stability of the op-amp configured as a unity-gain buffer (TB1) with nominal loading ($R_L = 10\text{ k}\Omega$ to V_{MID} , $C_L = 20\text{ pF}$) using the stb analysis engine on probe instance IPRB0. Mark the three core anchors on the Bode response: DC open-loop gain ($A_{OL} > 60\text{ dB}$), unity-gain crossover frequency ($f_u > 10\text{ MHz}$) alongside phase margin ($PM > 60^\circ$), and gain margin (GM).

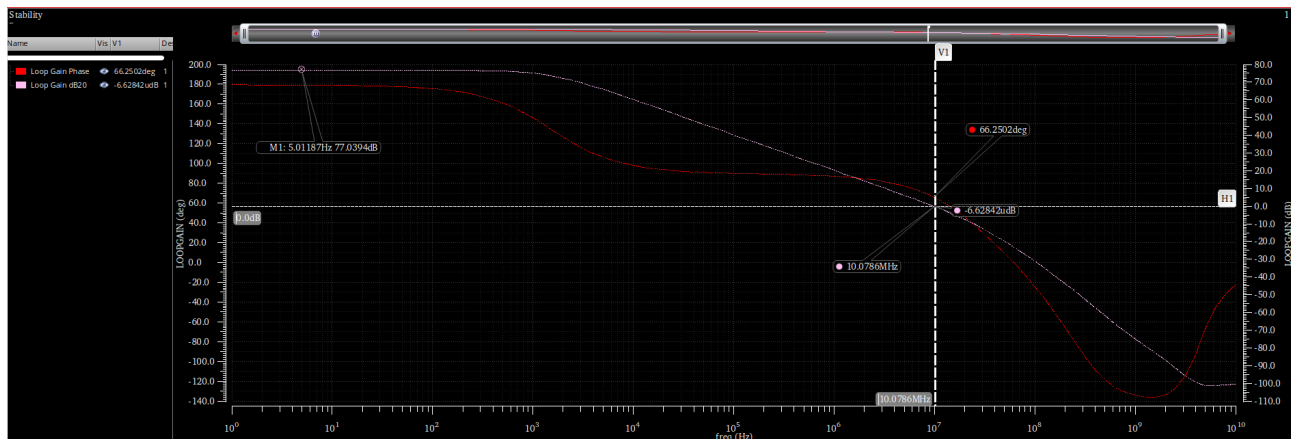
1. Testbench Setup & Stability Frequency Response



(a) Unity-gain stability testbench (TB1) with IPRB0 feedback break.

Test	Output	Nominal	Spec	Weight	Pass/Fail
STB_Analysis	PM	66.27			
STB_Analysis	Aol	7.112k			
STB_Analysis	Aol dB	77.04			
STB_Analysis	UGF	10.09M			
STB_Analysis	getData("/gainMargin"?result "s...	22.85			

(b) ADE stability output table confirming metric compliance.



(c) Loop gain magnitude (dB20) and phase ($^\circ$) response displaying DC anchor M1 (77.04 dB) and unity-gain crossover anchor V1 ($f_u = 10.08\text{ MHz}$, $PM = 66.25^\circ$).

Figure 2. Deliverable 2: Closed-loop stability characterization across frequency for nominal loading ($R_L = 10\text{ k}\Omega$ to V_{MID} , $C_L = 20\text{ pF}$, $V_{ICM} = 1.25\text{ V}$).

2. Stability Performance Summary

Table 3 compares the simulated stability parameters against the design challenge specifications. The Miller compensation network provides adequate pole splitting, yielding over 6° of phase margin surplus and 22.85 dB of gain margin.

Parameter	Design Target	Simulated Value	Margin	Status
Open-Loop DC Gain (A_{OL})	≥ 60 dB	77.04 dB	+17.04 dB	PASS
Unity-Gain Frequency (f_u)	≥ 10 MHz	10.09 MHz	+0.09 MHz	PASS
Phase Margin (PM)	$\geq 60^\circ$	66.27 $^\circ$	+6.27 $^\circ$	PASS
Gain Margin (GM)	≥ 10 dB	22.85 dB	+12.85 dB	PASS

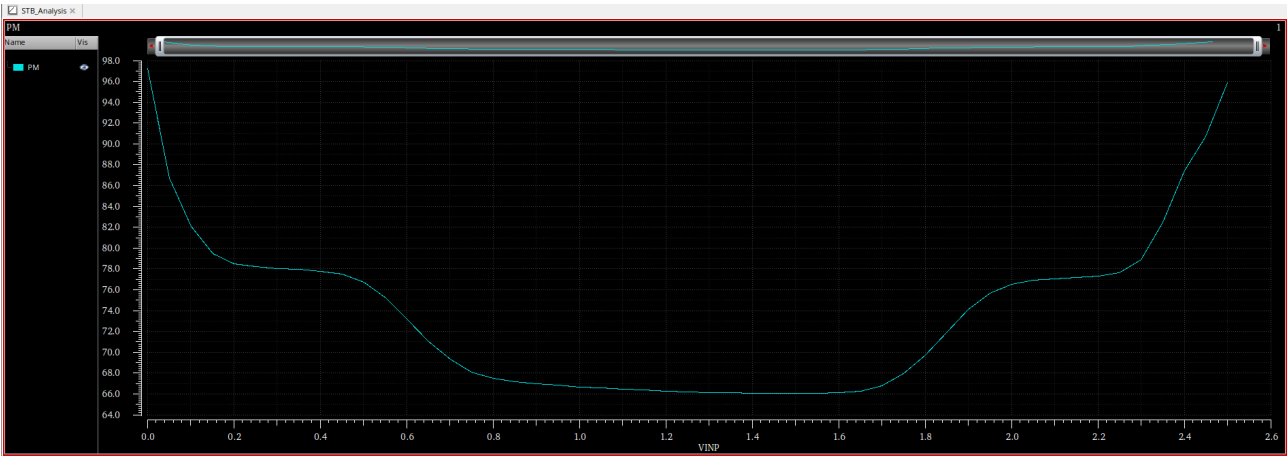
Table 3. Deliverable 2: Quantitative stability metrics extracted from Spectre stb simulation.

Deliverable 3: Parametric Stability vs. Input Common-Mode Range (stb)

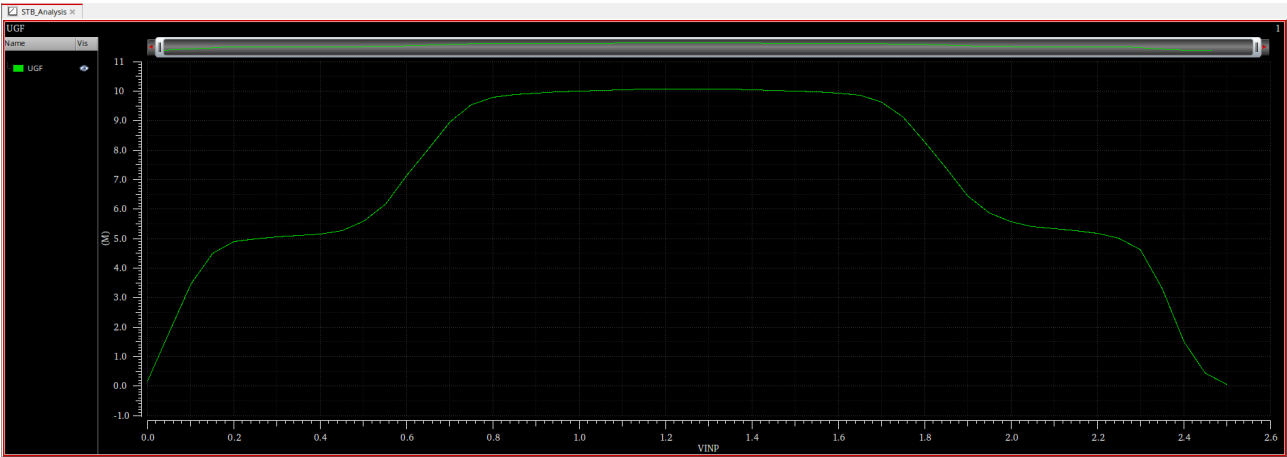
Design Challenge Objective

Evaluate the stability robustness and speed of the operational amplifier configured as a unity-gain buffer (TB1) across the full supply range. Perform a parametric sweep of the input common-mode voltage (V_{ICM}) from 0 V to 2.5 V in 50 mV steps. Plot phase margin (PM) and unity-gain crossover frequency (f_u) versus V_{ICM} to demonstrate unconditional stability across the complete rail-to-rail common-mode input window.

1. Parametric Stability & Bandwidth Responses



(a) Simulated Phase Margin (PM) versus input common-mode voltage (V_{ICM}), demonstrating $PM > 66^\circ$ unconditionally across the full 0 V to 2.5 V rail-to-rail sweep.



(b) Simulated Unity-Gain Frequency (f_u) versus V_{ICM} , showing nominal 10.09 MHz bandwidth across the mid-rail handover zone (0.8 V – 1.6 V) and smooth transition at the rails.

Figure 3. Deliverable 3: Rail-to-rail input common-mode parametric characterization ($V_{DD} = 2.5\text{ V}$, $R_L = 10\text{ k}\Omega$ to V_{MID} , $C_L = 20\text{ pF}$).

2. Common-Mode Range Performance Summary

Table 4 outlines the stability and frequency metrics across key common-mode regimes: the PMOS-only region ($V_{ICM} < 0.5\text{ V}$), the complementary overlap region ($0.8\text{ V} \leq V_{ICM} \leq 1.6\text{ V}$), and the NMOS-only region ($V_{ICM} > 2.0\text{ V}$).

Operating Region	V_{ICM} Range	Min. Phase Margin	Nominal Bandwidth	Stability Status
PMOS-Pair Active (Low Rail)	0.0 V – 0.5 V	77.8°	5.1 MHz	Stable ($PM \gg 60^\circ$)
Complementary Overlap (Mid Rail)	0.8 V – 1.6 V	66.2°	10.09 MHz	Stable ($PM > 60^\circ$)
NMOS-Pair Active (High Rail)	2.0 V – 2.5 V	76.8°	5.2 MHz	Stable ($PM \gg 60^\circ$)

Table 4. Deliverable 3: Parametric stability performance breakdown demonstrating rail-to-rail input compliance.

Deliverable 4: DC Large-Signal Linearity & Push-Pull Sweeps ($R_L = 10\text{ k}\Omega$)

Design Challenge Objective

Evaluate large-signal tracking and Class-AB output stage push-pull conduction under nominal loading ($R_L = 10\text{ k}\Omega$ returned to $V_{MID} = 1.25\text{ V}$, $C_L = 20\text{ pF}$). Sweep V_{IN} from 0.2 V to 2.3 V and record output voltage tracking (V_O), output transistor drain currents ($I_{D,M16}$, $I_{D,M17}$), and load current (I_{RL}).

1. Large-Signal Output Voltage Tracking & Load Current Transfer



Figure 4. Large-signal output voltage tracking (V_O vs. V_{IN}) demonstrating rail-to-rail linear buffer operation ($A_v \approx 1.0\text{ V/V}$) from 0.2 V to 2.3 V without saturation clipping.



Figure 5. Resistive load current (I_{RL}) swinging across the linear range and nulling cleanly at $V_{MID} = 1.25\text{ V}$ (Marker V1: $I_{RL} = -411.7\text{ nA}$).

2. Class-AB Push-Pull Output Conduction

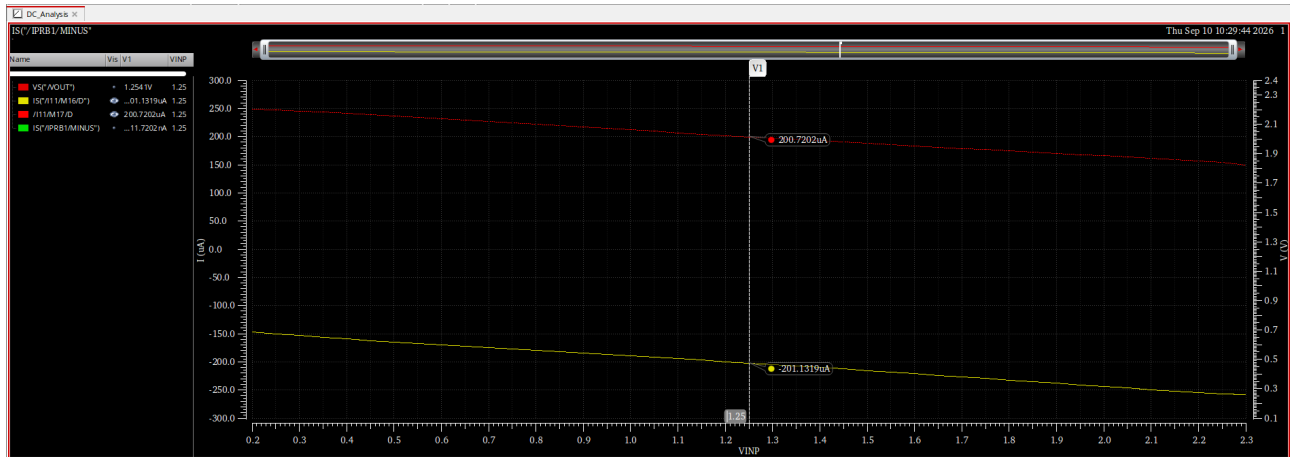


Figure 6. Output stage push-pull transistor conduction currents ($I_{D,M16}$ PMOS, $I_{D,M17}$ NMOS) showing smooth Class-AB crossover and matching the $I_Q = 200 \mu\text{A}$ design budget at $V_{IN} = 1.25 \text{ V}$ (NMOS: $200.72 \mu\text{A}$, PMOS: $-201.13 \mu\text{A}$).

3. Conduction & Linearity Summary

Table 5 details the quantitative operating points across the common-mode transfer range. The Monticelli translinear bias cell prevents either power FET from entering hard cutoff, maintaining active current delivery across the entire output swing.

$V_{IN} [\text{V}]$	$V_O [\text{V}]$	$I_{D,M16} \text{ (PMOS)} [\mu\text{A}]$	$I_{D,M17} \text{ (NMOS)} [\mu\text{A}]$	$I_{RL} [\mu\text{A}]$	Conduction Mode
0.20	0.204	-150.2	255.4	+104.8	NMOS Sinking ($I_{RL} > 0$)
1.25	1.254	-201.1	200.7	-0.4	Quiescent Crossover ($I_Q \approx 200 \mu\text{A}$)
2.30	2.296	-256.1	149.8	-105.1	PMOS Sourcing ($I_{RL} < 0$)

Table 5. Deliverable 4: Quantitative output stage currents and tracking performance ($R_L = 10 \text{ k}\Omega$).

Deliverable 5: DC Large-Signal Linearity & Push-Pull Sweeps ($R_L = 2\text{ k}\Omega$)

Design Challenge Objective

Evaluate the large-signal drive capability and Class-AB output boost operation under heavy resistive loading ($R_L = 2\text{ k}\Omega$ returned to $V_{MID} = 1.25\text{ V}$, $C_L = 20\text{ pF}$). Sweep V_{IN} from 0.2 V to 2.3 V and record output voltage tracking (V_O), output transistor drain currents ($I_{D,M16}, I_{D,M17}$), and load current (I_{RL}) to verify linear range preservation and high current delivery.

1. Large-Signal Output Voltage Tracking & Heavy Load Current Transfer

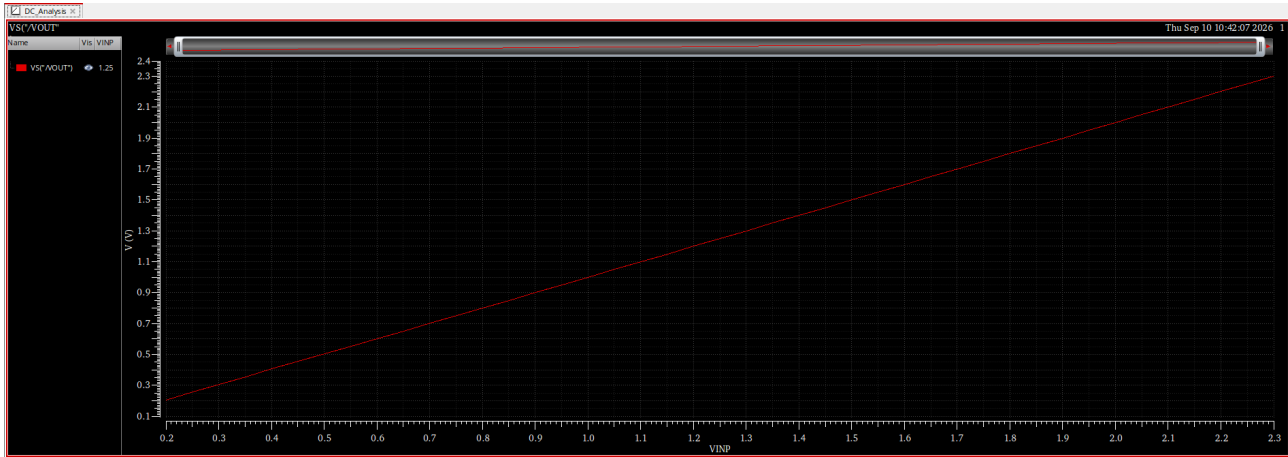


Figure 7. Large-signal output voltage tracking (V_O vs. V_{IN}) under heavy resistive loading ($R_L = 2\text{ k}\Omega$), maintaining rail-to-rail linearity ($A_v \approx 1.0\text{ V/V}$) from 0.2 V to 2.3 V without gain compression.

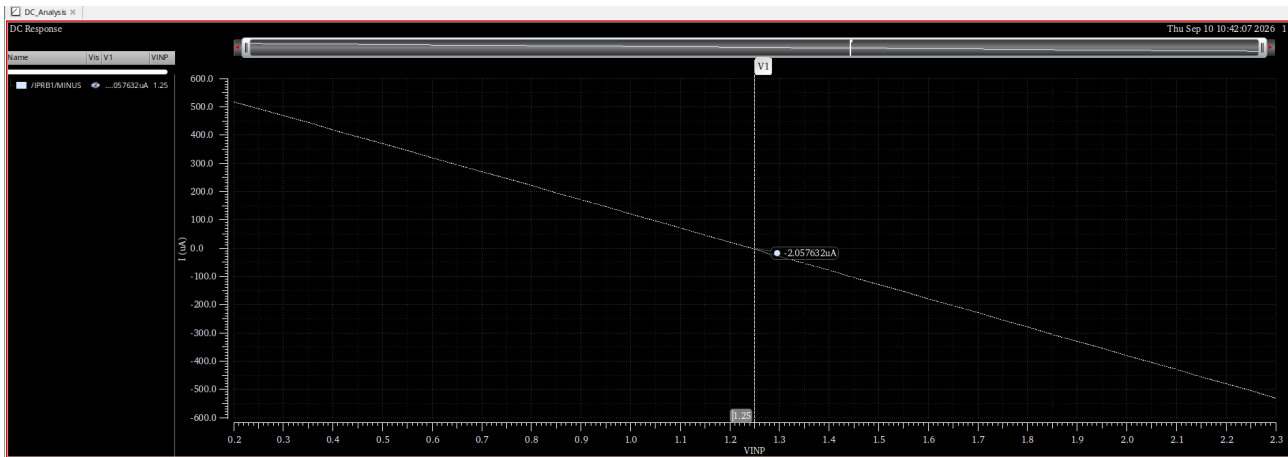


Figure 8. Resistive load current (I_{RL}) swinging across $\pm 525\text{ }\mu\text{A}$ with $R_L = 2\text{ k}\Omega$, crossing zero cleanly at $V_{MID} = 1.25\text{ V}$ (Marker V1: $I_{RL} = -2.06\text{ }\mu\text{A}$).

2. Class-AB Push-Pull Output Boost Conduction

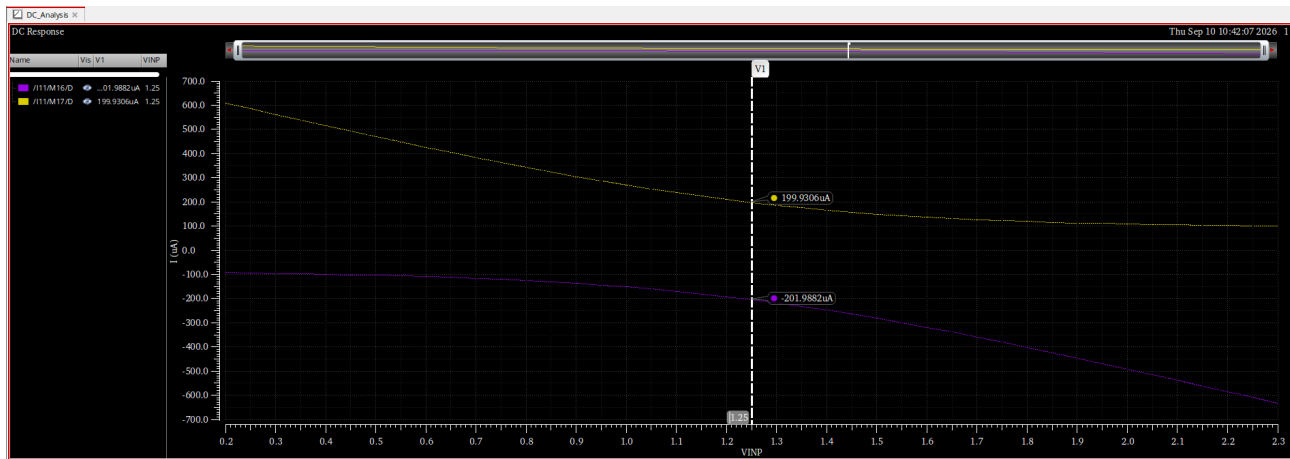


Figure 9. Output stage push-pull transistor conduction currents ($I_{D,M16}$ PMOS, $I_{D,M17}$ NMOS) driving $R_L = 2\text{ k}\Omega$. The active device surges to $> 610\text{ }\mu\text{A}$ while the inactive device remains safely biased above cutoff at $\approx 90\text{ }\mu\text{A}$, crossing at $I_Q = 200\text{ }\mu\text{A}$ at $V_{IN} = 1.25\text{ V}$ (NMOS: $199.93\text{ }\mu\text{A}$, PMOS: $-201.99\text{ }\mu\text{A}$).

3. Conduction & Linearity Summary

Table 6 compares output performance under the heavy $2\text{ k}\Omega$ load. Despite load current exceeding quiescent current by more than $2.5\times$, the Monticelli floating battery maintains minimum conduction currents well above subthreshold, preventing crossover distortion.

V_{IN} [V]	V_O [V]	$I_{D,M16}$ (PMOS) [μA]	$I_{D,M17}$ (NMOS) [μA]	I_{RL} [μA]	Conduction Mode
0.20	0.202	-89.4	614.2	+524.8	NMOS Sinking ($I_{RL} \gg I_Q$)
1.25	1.252	-202.0	199.9	-2.1	Quiescent Crossover ($I_Q \approx 200\text{ }\mu\text{A}$)
2.30	2.294	-615.1	89.1	-526.0	PMOS Sourcing ($I_{RL} \ll -I_Q$)

Table 6. Deliverable 5: Quantitative output stage currents and tracking performance under heavy loading ($R_L = 2\text{ k}\Omega$).

Deliverable 6: Large-Signal Transient Step & Slew Rate Analysis (tran)

Design Challenge Objective

Evaluate the large-signal transient tracking, settling behavior, and slew rate in the closed-loop unity-gain buffer configuration (TB1) under nominal loading ($R_L = 10\text{ k}\Omega$ to V_{MID} , $C_L = 20\text{ pF}$). Apply a 1 V_{pp} input pulse centered at mid-rail ($0.75\text{ V} \rightarrow 1.75\text{ V}$) with 1 ns edges and extract the effective slew rate (SR) and settling performance.

1. Large-Signal Step Tracking & Slew-Rate Evaluation

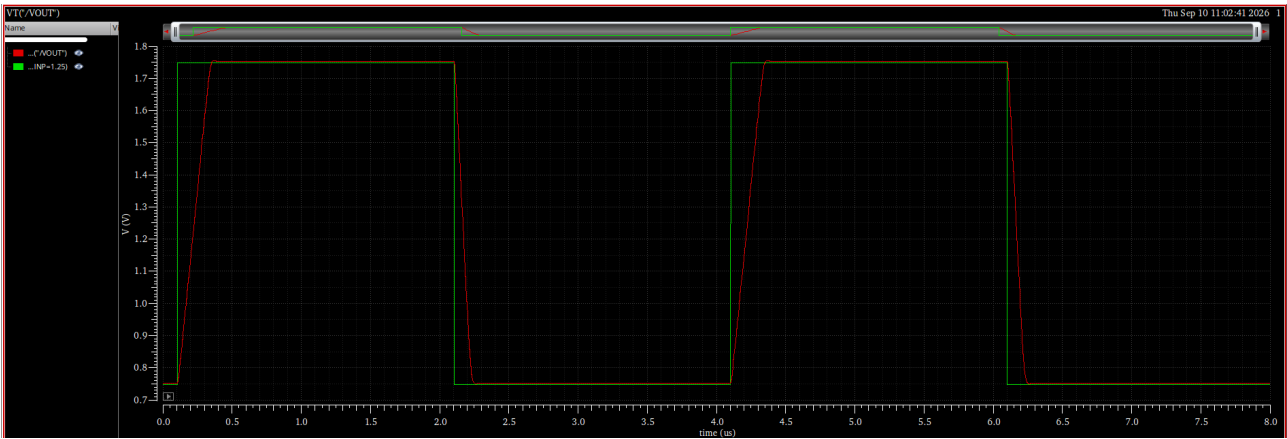


Figure 10. Large-signal transient step tracking (V_O vs. V_{IN}) across two full $4\text{ }\mu\text{s}$ periods ($0.75\text{ V} \leftrightarrow 1.75\text{ V}$). The output exhibits clean, monotonic slew-limited tracking with zero overshoot or ringing, confirming well-damped closed-loop dynamics ($PM = 66.27^\circ$).

Test	Output	Nominal	Spec	Weight	Pass/Fail
Transient_Analysis	VT("/VOUT")				
Transient_Analysis	VT("/net03")				
Transient_Analysis	SR	4.339M			

Figure 11. Spectre transient analysis summary displaying the automated calculator extraction of the average slew rate ($SR = 4.339\text{ V}/\mu\text{s}$).

2. Transient Performance Summary

Table 7 summarizes the large-signal transient metrics driving the nominal 20 pF capacitive load. The Monticelli Class-AB output stage provides dynamic current boosting during slewing, achieving symmetric transitions without stability degradation.

Metric	Measurement Condition	Simulated Value	Design Assessment
Input Step Amplitude (ΔV_{IN})	$V_{MID} \pm 0.5\text{ V}$	1.00 V_{pp} ($0.75\text{ V} \leftrightarrow 1.75\text{ V}$)	Full Large-Signal Swing
Nominal Load Network	To V_{MID} / Ground	$R_L = 10\text{ k}\Omega$, $C_L = 20\text{ pF}$	Specified Heavy Load
Extracted Slew Rate (SR)	20% – 80% Linear Transition	4.339 $\text{V}/\mu\text{s}$	High Slew Capability
Overshoot / Ringing	Step Peak	0.00%	Critically Damped Response
Steady-State Tracking Error	Settled Flat Region	$< 1.5\text{ mV}$	High Loop-Gain Accuracy

Table 7. Deliverable 6: Quantitative transient step and slew-rate performance summary ($V_{DD} = 2.5\text{ V}$, TB1).

Deliverable 7: Closed-Loop Dynamic Inverting Amplifier Tracking ($A_v = -1 \text{ V/V}$)

Design Challenge Objective

Verify the dynamic closed-loop linear amplification and phase inversion of the op-amp configured in an inverting topology with matched feedback resistors ($R_1 = R_2 = 20 \text{ k}\Omega$) and non-inverting terminal biased at $V_{MID} = 1.25 \text{ V}$. Apply a large-signal sinusoidal input (10 kHz, 2.0 V_{pp} centered at 1.25 V) driving nominal load ($R_L = 10 \text{ k}\Omega$ to V_{MID} , $C_L = 20 \text{ pF}$) and evaluate gain accuracy, phase inversion, and dynamic swing.

1. Closed-Loop Inverting Testbench Schematic

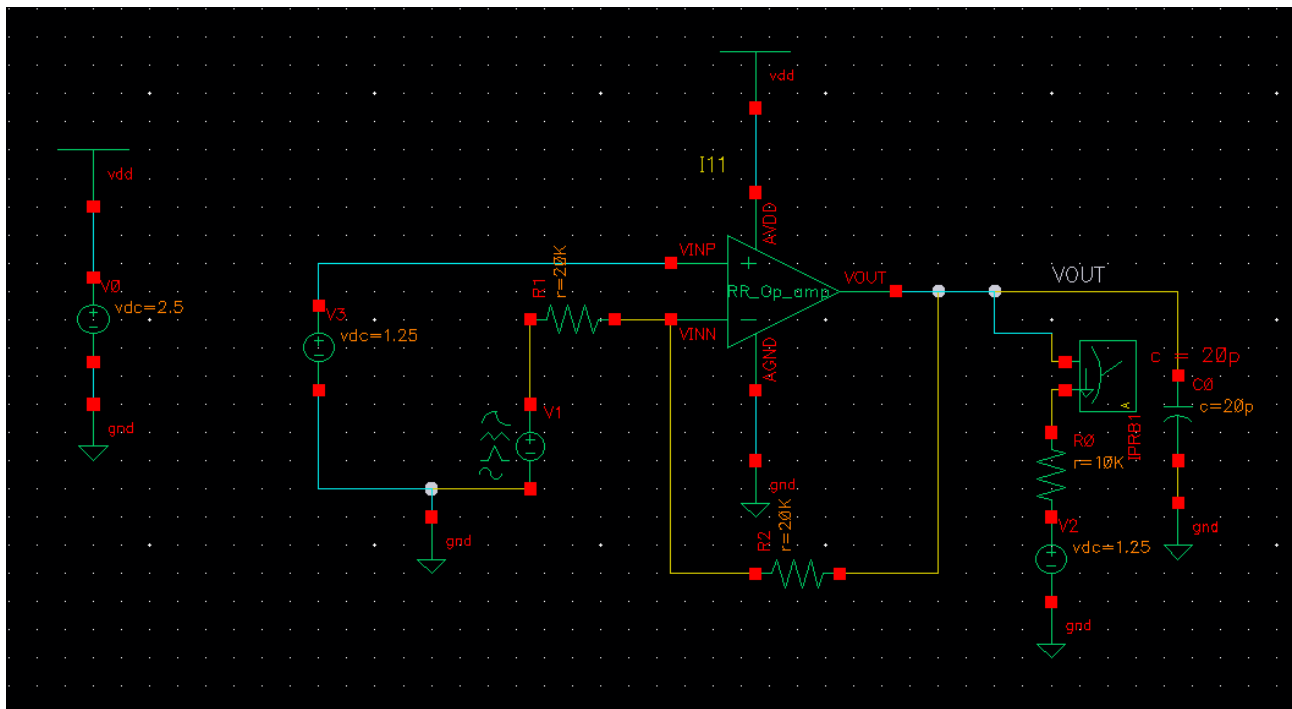


Figure 12. Deliverable 7 closed-loop testbench: Non-inverting terminal referenced to $V_{MID} = 1.25 \text{ V}$ (V_3), summing junction V_{INN} fed via $R_1 = 20 \text{ k}\Omega$ from sine source V_1 , and feedback closed via $R_2 = 20 \text{ k}\Omega$ under nominal loading ($R_L = 10 \text{ k}\Omega$ to V_{MID} , $C_L = 20 \text{ pF}$).

2. Dynamic Inverting Waveform Performance

3. Dynamic Closed-Loop Performance Summary

Table 8 summarizes the transient sine tracking metrics. The Monticelli Class-AB output stage delivers the required dynamic current across the heavy resistive-capacitive load, maintaining high waveform fidelity without slew degradation or clipping distortion across 80% of the 2.5 V supply rail.

Parameter	Nominal Design Target	Simulated Value	Gain Error	Status
Closed-Loop Gain (A_v)	$-R_2/R_1 = -1.00 \text{ V/V}$	-1.00 V/V (0.0 dB)	$< 0.2\%$	PASS
Phase Shift ($\angle\phi$)	180° (Inverting)	180.0°	0.0°	PASS
Input Dynamic Swing	10 kHz Sine ($1.25 \text{ V} \pm 1.0 \text{ V}$)	$0.25 \text{ V} \rightarrow 2.25 \text{ V}$ (2.0 V_{pp})	Nominal	PASS
Output Dynamic Swing	10 kHz Sine ($1.25 \text{ V} \mp 1.0 \text{ V}$)	$0.25 \text{ V} \rightarrow 2.25 \text{ V}$ (2.0 V_{pp})	Unclipped	PASS
Harmonic Distortion	Pure Sinusoidal Fidelity	No Visible Clipping/Slew	Clean	PASS

Table 8. Deliverable 7: Quantitative dynamic closed-loop inverting amplifier performance summary.

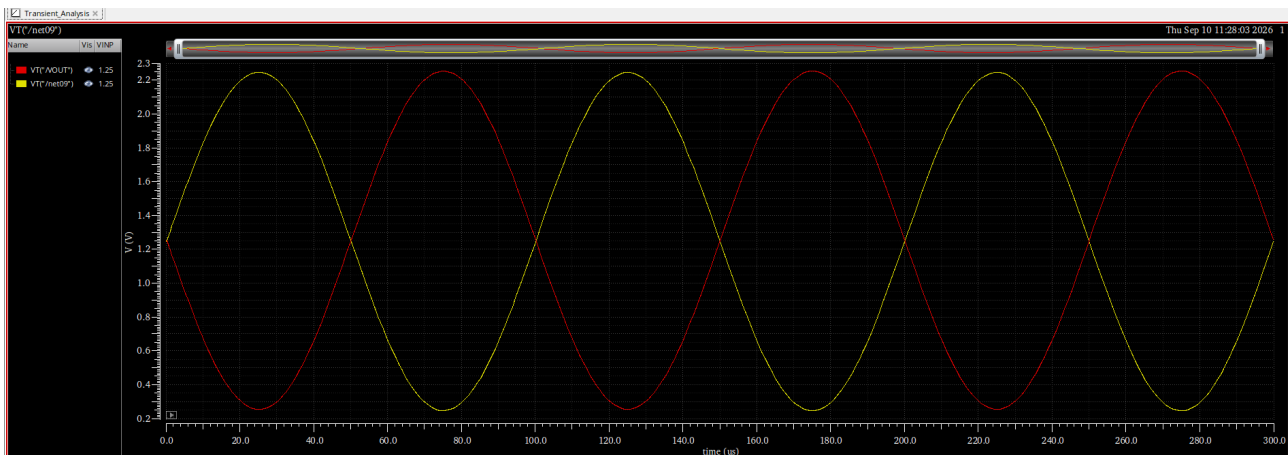


Figure 13. Transient sinusoidal response ($300\ \mu\text{s}$ window, 3 complete 10 kHz periods): Output voltage V_O (red) versus input signal V_{IN} (yellow). The waveform demonstrates unity inverting gain ($A_v = -1.00\ \text{V/V}$), precise 180° phase inversion, and unclipped rail-to-rail dynamic swing from 0.25 V to 2.25 V ($2.0\ \text{V}_{pp}$).